FACTORIZATION OF THE ZERO SCHEME OF A SECTION OF A TENSOR PRODUCT OF INVERTIBLE SHEAVES OVER A FLAT FAMILY OF RELATIVE DIMENSION ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted proof opens with a displayed commutative diagram in Xy-pic, which is why the native xy package of the shared Stacks preamble is retained; the diagram is the author's own and was not redrawn.

Allowed standard prerequisites (editorial).

- Relative spectrum of a quasi-coherent sheaf of algebras. Same chapter, chow.tex, Section 'Relative spectrum' (Stacks tag 01LQ, label section-relative-spectrum). Identifies the relative spectrum of the graded quotient algebra A as the flat family p : X -> Y carrying the two sections s_i; the actual construction of A and of the adjoint sections is inside the counted span.
- Zero scheme of a global section of an invertible module. Divisors, tag 01WX (label lemma-section-and-zero-scheme) and Schemes, tag 01VZ on closed subschemes cut out by an ideal sheaf. Supplies that D_i, D and the closed subschemes cut out by the local equations z_i are the zero schemes of the corresponding sections; the identification of D_i with the Spec of Sym of the dual invertible module is carried out in the counted span.

Lemma 0.1. *Let Y be a scheme. Let $\mathcal{L}_i$, $i = 1, 2$ be invertible $\mathcal{O}_Y$-modules. Let s be a global section of $\mathcal{L}_1 \otimes_{\mathcal{O}_Y} \mathcal{L}_2$. Denote $i : D \to Y$ the zero scheme of s. Then there exists a commutative diagram*

$$\begin{array}{ccccc} D_1 & \xrightarrow[i_1]{} & L & \xleftarrow[i_2]{} & D_2 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p_2} \\ D & \xrightarrow{i} & Y & \xleftarrow{i} & D \end{array}$$

and sections s_i of $p^*\mathcal{L}_i$ such that the following hold:

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(1) $p^*s = s_1 \otimes s_2$,
(2) p *is of finite type and flat of relative dimension* 1,
(3) D_i *is the zero scheme of* s_i,
(4) $D_i \cong \underline{\mathrm{Spec}}(Sym^*(\mathcal{L}_{3-i}^{\otimes -1})|_D))$ *over* D *for* $i = 1, 2$,
(5) $p^{-1}D = D_1 \cup D_2$ *(scheme theoretic union),*
(6) $D_1 \cap D_2$ *(scheme theoretic intersection) maps isomorphically to* D, *and*
(7) $D_1 \cap D_2 \to D_i$ *is the zero section of the line bundle* $D_i \to D$ *for* $i = 1, 2$.

Moreover, the formation of this diagram and the sections s_i *commutes with arbitrary base change.*

Proof. Let $p : X \to Y$ be the relative spectrum of the quasi-coherent sheaf of $\mathcal{O}_Y$-algebras

$$\mathcal{A} = \left(\bigoplus\nolimits_{a_1, a_2 \geq 0} \mathcal{L}_1^{\otimes -a_1} \otimes_{\mathcal{O}_Y} \mathcal{L}_2^{\otimes -a_2} \right) / \mathcal{J}$$

where $\mathcal{J}$ is the ideal generated by local sections of the form $st - t$ for t a local section of any summand $\mathcal{L}_1^{\otimes -a_1} \otimes \mathcal{L}_2^{\otimes -a_2}$ with $a_1, a_2 > 0$. The sections s_i viewed as maps $p^*\mathcal{L}_i^{\otimes -1} \to \mathcal{O}_X$ are defined as the adjoints of the maps $\mathcal{L}_i^{\otimes -1} \to \mathcal{A} = p_*\mathcal{O}_X$. For any $y \in Y$ we can choose an affine open $V \subset Y$, say $V = \mathrm{Spec}(A)$, containing y and trivializations $z_i : \mathcal{O}_V \to \mathcal{L}_i^{\otimes -1}|_V$. Observe that $f = s(z_1 z_2) \in A$ cuts out the closed subscheme D. Then clearly

$$p^{-1}(V) = \mathrm{Spec}(B[z_1, z_2]/(z_1 z_2 - f))$$

Since D_i is cut out by z_i everything is clear. □